Formula for finding square root of a complex number

$$\sqrt{a+ib}=\pm\left(\sqrt{\frac{|z|+a}{2}}+i\frac{b}{|b|}\sqrt{\frac{|z|-a}{2}}\right),\ \text{where}\ z=a+ib\ \text{and}\ b\neq 0.$$

Let r and θ be polar coordinates of the point $P(x,y)$ that corresponds to a non-zero complex number $z=x+iy$. The polar form or trigonometric form of a complex number P is

$$z=r(\cos\theta+i\sin\theta).$$

Properties of polar form

Property 1: If $z=r(\cos\theta+i\sin\theta)$, then $z^{-1}=\frac{1}{r}(\cos\theta-i\sin\theta)$.

Property 2: If $z_1=r_1(\cos\theta_1+i\sin\theta_1)$ and $z_2=r_2(\cos\theta_2+i\sin\theta_2)$, then $z_1z_2=r_1r_2(\cos(\theta_1+\theta_2)+i\sin(\theta_1+\theta_2))$.

Property3: If $z_1=r_1(\cos\theta_1+i\sin\theta_1)$ and $z_2=r_2(\cos\theta_2+i\sin\theta_2)$, then $\frac{z_1}{z_2}=\frac{r_1}{r_2}\left[\cos(\theta_1-\theta_2)+i\sin(\theta_1-\theta_2)\right]$.

de Moivre's Theorem

(a) Given any complex number $\cos\theta+i\sin\theta$ and any integer n,

$$(\cos\theta+i\sin\theta)^n=\cos n\theta+i\sin n\theta$$

(b) If x is rational, then $\cos x\theta+i\sin x\theta$ in one of the values of $(\cos\theta+i\sin\theta)^x$

The nth roots of complex number $z=r(\cos\theta+i\sin\theta)$ are

$$z^{1/n}=r^{1/n}\left(\cos\left(\frac{\theta+2k\pi}{n}\right)+i\sin\left(\frac{\theta+2k\pi}{n}\right)\right),\ k=0,1,2,3,\ldots,n-1.$$

ICT CORNER

https://ggbm.at/vchq92pg or Scan the QR Code

Open the Browser, type the URL Link given below (or) Scan the QR code. GeoGebra work book named "12th Standard Mathematics" will open. In the left side of the work book there are many chapters related to your text book. Click on the chapter named "Complex Numbers". You can see several work sheets related to the chapter. Select the work sheet "Geometrical Meaning"

CPL9J

12th_Maths_EM_Vol1_CH 2_Complex Numbers.indd 96 16-12-2022 12:33:36